

Thomas Brazelton

Harvard University [tbrazel.github.io](https://github.com/tbrazel) tbraz@alumni.upenn.edu

- EMPLOYMENT **Vanderbilt University** Nashville, TN
Assistant Professor Spring 2027 —
- Simons Laufer Mathematical Sciences Institute** Berkeley, CA
Uhlenbeck Postdoctoral Fellow Fall 2026
Motivic Homotopy Theory: Connections and Applications
- Harvard University** Cambridge, MA
NSF Postdoctoral Research Fellow 2023—2026
Sponsor: Mike Hopkins
- EDUCATION **University of Pennsylvania** Philadelphia, PA
Ph.D, Mathematics 2018—2023
NSF Graduate Research Fellow
Advisors: Mona Merling & Kirsten Wickelgren
- The Johns Hopkins University** Baltimore, MD
B.A. + M.A., Mathematics 2014—2018
- PREPRINTS 20. Operations on G -sets and the Dress map, with B. Spitz. 18 pages, 2026.
 19. The evolution of enumerative geometry: a narrative from classical problems to enriched invariants, with C. Bethea. 23 pages, 2025.
 18. On rank 2 bundles over smooth affine fourfolds, with M. Opie, T. Syed. 37 pages, 2025.
 17. The Chow–Witt rings of the classifying spaces of quadratically oriented bundles, with M. Wendt. 43 pages, 2025.
 16. Monodromy in the space of symmetric cubic surfaces with a line, with S. Raman. 28 pages, 2024.
- PUBLICATIONS 15. C_p -Mackey functors in Macaulay2, with D. Chan, B. Mudrak, B. Spitz, C. Vogeli, C. Wang, M. Zeng, S. Zotine. *J. Software Alg. Geom.* (2026).
 14. Bitangents to symmetric quartics, with C. Bethea. *J. Alg.* (2026).
 13. Concerning monoid structures on naive homotopy classes of endomorphisms of punctured affine space, with W. Hornslien. *J. Homotopy Relat. Struc.* (2025).
 12. An enriched degree of the Wronski map, *New York Journal of Mathematics* **31** (2025), 195—222.
 11. Equivariant enumerative geometry. *Advances in Mathematics* **461**, 2025.
 10. $\mathbb{A}^1$ -Brouwer degrees in Macaulay2, with N. Borisov, F. Espino, T. Hagedorn, Z. Han, J. Lopez Garcia, J. Louwsma, G. Ong, and A. Tawfeek. *J. Software Alg. Geom.* **14** (2024) pp. 175–187.
 9. Residue sums of Dickson polynomials over finite fields, with J. Harrington, M. Litman, T. H. W. Wong. *J. Number Theory* **264** (2024) pp. 1—26.
 8. Lifts, transfers, and degrees of univariate maps, with S. McKean. *Mathematica Scandinavica* **129** (2023) pp. 5—38.
 7. Bézoutians and the $\mathbb{A}^1$ -degree, with S. McKean and S. Pauli *Algebra & Number Theory* **17(11)**, 2023.
 6. Homotopy Mackey functors of equivariant algebraic K -theory. *Journal of Pure and Applied Algebra*, **226(8)**, August 2022.
 5. An introduction to $\mathbb{A}^1$ -enumerative geometry. In Neumann F., Pál A. (eds) *Homotopy Theory and Arithmetic Geometry — Motivic and Diophantine Aspects*. Lecture notes in Mathematics, vol **2292**. Springer, Cham. 2021.
 4. A note on semilinearization and twisted group rings. *Communications in Algebra*, **49:8**, 3380—3386, 2021.
 3. The trace of the local $\mathbb{A}^1$ -degree, with R. Burklund, M. McKean, M. Montoro, M. Opie. *Homology, Homotopy and Applications* **23(1)** (2021) pp. 1—14.
 2. Zeros of newform Eisenstein series on $\Gamma_0(N)$, with V. Jakicic, *J. Number Theory* **190** (2018) pp. 109—130.

	1. On consecutive primitive n th roots of unity modulo q , with J. Harrington, S. Kannan, and M. Litman, <i>J. Number Theory</i> 174 (2017) pp. 494—504.	
SOFTWARE	MackeyFunctors.jl, a Julia package for Mackey functors for arbitrary finite groups. With S. Atherton, M. Franz, D. Isaksen, C. Vogeli, L. Zhou. CpMackeyFunctors, a Macaulay2 package for doing homological algebra with C_p-Mackey functors. With D. Chan, B. Mudrak, B. Spitz, C. Vogeli, C. Wang, M. Zeng, and S. Zotine. A1BrouwerDegrees, a Macaulay2 package for $\mathbb{A}^1$-Brouwer degree computation and working with symmetric bilinear forms. With N. Borisov, F. Espino, T. Hagedorn, Z. Han, J. Lopez Garcia, J. Louwsma, G. Ong, and A. Tawfeek.	
PROFESSIONAL SERVICE	<i>Project leader</i>, Machine Computation in Homotopy Theory, ICERM 2026 <i>Co-Organizer</i>, Macaulay2 Workshop 2025, UW-Madison (DMS-2508868) <i>Co-Organizer</i>, Mid-Atlantic Topology Conference 2024, Northeastern University (DMS-2349755) <i>Co-Organizer</i>, AMS Special Session on Homotopy Theory, JMM 2024 <i>Project leader</i>, Macaulay2 Workshop 2023, University of Minnesota	
COMMITTEE WORK	Graduate Admissions Committee, Harvard Mathematics, 2024–2026 Community Committee, Harvard Mathematics, 2024–2026	
REFeree WORK	<i>Referee</i>: Advances in Mathematics, Annali della Scuola Normale Superiore di Pisa, Geometry & Topology, IMRN <i>Reviewer</i>, MathSciNet	
RESEARCH TALKS	Machine Computation in Homotopy Theory, ICERM	Jun 2026
	Cornell Topology Festival	May 2026
	AMS-Sectional: Applied & Computational Algebraic Geometry Session, Savannah	Mar 2026
	BU Geometry & Physics Seminar	Feb 2026
	eCHT Research Seminar, Online	Jan 2026
	No boundaries seminar, University of Chicago	Nov 2025
	Colloquium, Université de Montréal	Oct 2025
	Boston College Algebraic Geometry Seminar	Sep 2025
	SIAM AG Minisymposium: Numerical and certified methods in algebraic geometry	Jul 2025
	Arithmetic, algebraic K -theory and algebraic cycles, OSU	May 2025
	UPenn Geometry & Topology Seminar	Mar 2025
	AMTRaK Seminar, UVA	Feb 2025
	UVA Topology Seminar	Feb 2025
	Brown Algebra Seminar	Jan 2025
	Notre Dame Topology Seminar	Jan 2025
	UW-Madison Algebra and Algebraic Geometry Seminar	Dec 2024
	CU-Boulder Topology Seminar	Nov 2024
	Harvard/MIT Algebraic Geometry Seminar	Oct 2024
	Enumerative Geometry Beyond Spaces, Banff	Aug 2024
	Workshop on Applied and Computational Enumerative Geometry, Fields Institute	Jun 2024
	Université du Québec à Montréal, LACIM Seminar	Apr 2024
	Algebra Seminar, University of Pennsylvania	Apr 2024
	Geometry/Topology Seminar, University of Pennsylvania	Apr 2024
	CIMAT Topology Seminar	Feb 2024
	University of Minnesota Topology Seminar	Feb 2024
	University of Chicago Algebraic Topology Seminar	Feb 2024
	University of Chicago No Boundaries Seminar	Feb 2024
	Northeastern Topology Seminar	Jan 2024
	UC-Irvine Topology Seminar	Jan 2024
	UCLA Topology Seminar	Jan 2024
	AMS Special Session on Equivariant Algebra, JMM	Jan 2024
	eCHT Research Seminar	Nov 2023
	Texas A&M Geometry Seminar	Nov 2023
	MIT Topology Seminar	Oct 2023
	Notre Dame Algebraic Geometry & Commutative Algebra Seminar	May 2023

University of Maryland Algebra & Number Theory Seminar	Mar 2023
Tulane University Algebraic Geometry & Geometric Topology Seminar	Feb 2023
Emory University Algebra Number Theory Seminar	Feb 2023
University of Waterloo Geometry/Topology Seminar	Feb 2023
University of Virginia Topology Seminar	Feb 2023
Brown University Algebraic Geometry Seminar	Feb 2023
JMM Special Session on Applied Enumerative Geometry	Jan 2023
UCLA Topology Seminar	Nov 2022
Rochester University Topology Seminar	Nov 2022
Binghamton University Graduate Conference (BUGCAT)	Nov 2022
Binghamton University Topology Seminar	Nov 2022
AMS Special Session on K -theory and Chromatic Homotopy Theory, U-Utah	Oct 2022
Johns Hopkins University Topology Seminar	Oct 2022
Loyola University Chicago Topology, Algebra, Combinatorics & Operators Seminar	Oct 2022
UChicago Topology Seminar	Oct 2022
Young Topologists Meeting 2022, Copenhagen	Jul 2022
Homotopy Theory with Applications to Arithmetic and Geometry, Fields Institute	Jun 2022
GROOT (Graduates Reminisce Online On Topology)	Jun 2022
Algebraic Structures in Topology, San Juan	Jun 2022
Algebra Seminar, Texas A&M University	Mar 2022
FRG Grant on Trace Methods	Jan 2022
SECANT 2021, Cedar Crest College	Jan 2022
Topology/Geometry Seminar, University of Oregon	Nov 2021
Geometry/Topology Seminar, University of Pennsylvania	Nov 2021
Algebra Seminar, University of Pennsylvania	Nov 2021
Young Topologists Meeting 2020/2021	Jul 2021
Algebra Seminar, University of Pennsylvania	Apr 2020
Binghamton University Graduate Conference (BUGCAT)	Nov 2020
Geometry/Topology Seminar, University of Pennsylvania	Oct 2020
AMS Fall Sectional, UW-Madison	Sep 2019
PIMS Workshop on Arithmetic Topology	Jun 2019
Rényi Institute Number Theory Seminar, Budapest	Sep 2016

AWARDS AND FELLOWSHIPS	AIM SQuaRE Grant	2023—2026,
	<i>Around the Wronski map</i> , with S. Karp, J. Levinson, S. McKean, K. Purbhoo, F. Sottile	
	AIM SQuaRE Grant	2023—2026,
	<i>An algebraic approach to Thom spectra</i> , with M. Calle, D. Chan, L. Keenan, M. Péroux	
	NSF Postdoctoral Research Fellowship	2023—2026
	AMS Travel Grant	2022
	Graduate Fellow for Teaching Excellence ,	
	Penn Center for Teaching and Learning	2022—2023
	Moez Alimohamed Graduate Student Award for Distinguished Teaching in Mathematics , University of Pennsylvania	2021
	Graduate Fellow for Equitable and Inclusive Teaching , Penn Center for Teaching and Learning	2021—2022
	Master TA , University of Pennsylvania Mathematics Department	2021—2023
	Dean's Scholar , University of Pennsylvania	
	Honored as one of nine doctoral students across the School of Arts & Sciences.	2020—2021
	Good Teaching Award , University of Pennsylvania Mathematics Department	Fall 2020
	NSF Graduate Research Fellow	2019 — 2024
	Calabi Fellow , University of Pennsylvania Mathematics Department	2017 — 2020
	J.J. Sylvester Award for Excellence in Mathematics (Johns Hopkins)	May 2018

	NSF GRFP , Honorable Mention	2018
	William Lowell Putnam Award (Johns Hopkins)	May 2016
TEACHING	Harvard University	
	MATH231br Topology of algebraic manifolds	Spring 2026
	MATH1B, Calculus	Fall 2025
	MATH101, Sets, groups, and geometry	Spring 2025
	MATH266, Unstable motivic homotopy theory	Fall 2024
	MIT Educational Justice Initiative	
	TA, Code your dreams (Python, HTML, Flask)	Spring 2025
	TA, Intro to Python	Fall 2024
	University of Pennsylvania	
	<i>Instructor</i> , MATH8100 Enumerative Geometry	Spring 2023
	Designed and taught an inquiry-based learning course on enumerative geometry for graduate students and advanced undergrads.	
	Penn Directed Reading Program	Fall 2019 — Spring 2023
	Co-founded and co-organized the DRP at Penn with Mona Merling (Fall 2019), co-organized with George Wang (Spring 2020 — Spring 2021), and with Marielle Ong (Fall 2021 — Spring 2023). Mentor for the following projects:	
	$\mathbb{A}^1$ -Milnor numbers, Zhong Zhang	Spring 2023
	<i>Algebraic geometry from an $\mathbb{A}^1$-viewpoint</i> , Zhong Zhang	Fall 2022
	<i>Enumerative geometry and string theory II</i> , Zhong Zhang	Spring 2022
	<i>Enumerative geometry and string theory</i> , Zhong Zhang	Fall 2021
	<i>Category theory and homotopy theory</i> , Abigail Timmel	Spring 2020
	<i>Group theory and applications</i> , Stephanie Wu	Fall 2019
	<i>Persistent homology</i> , Mira Wattal (JHU)	Spring 2018
	Princeton Prison Teaching Initiative	Spring 2021 — Spring 2022
	Volunteer math instructor for South Woods State Prison in New Jersey.	
	<i>Instructor/Team Leader for MATH015</i>	Spring 2022
	<i>Instructor for MATH020</i>	Fall 2021
	<i>Grader for MATH015</i>	Spring 2021
ADVISING	Penn Summer Prep	Philadelphia, PA
	<i>Instructor</i> , Introduction to Voting Theory	Summer 2021
	Designed and taught a two-week course on voting theory for advanced high school students.	
	University of Pennsylvania	Philadelphia, PA
	<i>Teaching Assistant</i> , MATH 370 Algebra I	Spring 2021
	<i>Teaching Assistant</i> , MATH 114 Calculus II	Fall 2020
	The Johns Hopkins University	Baltimore, MD
	<i>Teaching Assistant</i> , AS.110.421 Dynamical Systems	Spring 2018
	<i>Teaching Assistant</i> , AS.110.202 Calculus III	Fall 2017
	Senior thesis: Vinberg's algorithm (Charles Wolhgemuth)	2025-2026
	Senior thesis: automorphism groups of curves (David Ettel)	2025-2026
	Independent study: Schubert calculus and obstruction theory (Yuyuan Luo)	Spring 2024

SEMINARS ORGANIZED	<i>Harvard/MIT Juvitop</i> , The norm residue isomorphism theorem, (w/ Logan Hyslop)	Fall 2025
	<i>eCHT</i> ¹ , Quadratically enriched curve counting (w/ Sabrina Pauli)	Fall 2024
	<i>eCHT</i> , Algebraic, motivic, and topological vector bundles (w/ Morgan Opie)	Fall 2023
	<i>UPenn Homotopy Theory Seminar</i> (w/ Andres Mejia)	Spring 2021—Spring 2023
	<i>Penn Graduate Student Seminar</i> (w/ Marielle Ong)	Fall 2019—Spring 2020
FACILITATION	Center for Teaching and Learning, University of Pennsylvania	
	<i>Designing Problems for STEM Classes</i>	Fall 2022
	Designed a university-wide workshop on scaffolding and backwards design in problem sets in STEM.	
	<i>Inclusive and Equitable Teaching in STEM</i>	Spring 2022
	Designed and facilitated a five-session mini-course on inclusive and equitable teaching in STEM disciplines for graduate students.	
EXPOSITORY TALKS	<i>Inclusive and Equitable Teaching</i>	Fall 2021
	Co-facilitated a five-session mini-course on inclusive and equitable teaching for graduate students.	
	Texas A&M Undergrad Math Society	Nov 2023
	<i>Gimbal lock and covering spaces</i>	
	Penn Undergrad Math Society	Nov 2022
	<i>A hands-on introduction to homotopy</i>	
	Penn Geometry/Topology Grad Seminar	
	<i>Local homotopy theory and Galois descent</i>	Sep 2022
	<i>The algebraic vector bundle problem</i>	Feb 2022
	<i>Euler characteristics of real algebraic manifolds</i>	Oct 2021
	MIT Talbot Workshop, 2021	Oct 2021
	<i>Ambidexterity</i>	
	Penn General Robotics, Automation, Sensing & Perception Laboratory	Oct 2019
	<i>A brief introduction to topology</i>	
	Moravian College REU Seminar	Jun 2019
	<i>A hands-on introduction to homotopy</i>	
	University of Pennsylvania Graduate Pizza Seminar	
	<i>How to prevent nuclear war and then decide what to watch on Netflix</i>	Feb 2023
	<i>Elliptic curves and the NSA</i>	Jan 2020
	<i>Gimbal lock</i>	Dec 2020
	<i>Social choice and topology</i>	Mar 2019
	<i>The generalized Poincaré conjecture</i>	Oct 2018
	Johns Hopkins Undergraduate Mathematics Seminar	Apr 2018
	<i>A Crash Course in Homotopy Theory</i>	
SELECTED CONFERENCES AND WORKSHOPS (* INDICATES ONLINE)	Enumerative Geometry Beyond Spaces, Banff	Aug 2023
	PCMI Research Program in Motivic Homotopy Theory	Jul 2023
	International Workshop on Algebraic Topology, Shanghai	Jun 2024
	Algebraic Structures in Topology II, San Juan	Jun 2024
	Workshop on Computational and Applied Enumerative Geometry, Fields Institute	Jun 2024
	Motives in Mainz	Mar 2024
	Joint Mathematics Meetings 2024, San Francisco	Jan 2024
	European Autumn School in Topology 2023	Sep 2023
	Stacks Project Workshop 2023, Ann Arbor	Aug 2023
	Recent Advances in Algebraic K -Theory, IHES	Jul 2023
	Motivic and non-commutative aspects of enumerative geometry, Nijmegen	Jul 2023
	Scissors K -theory and Trace Methods, Indiana University	Jun 2023
	Macauley2 Workshop, University of Minnesota	May 2023
	Mid-Atlantic Topology Conference, Philadelphia	Apr 2023

¹electronic Computational Homotopy Theory seminar

Joint Mathematics Meeting, Boston	Jan 2023
Banff Workshop on Toric Degenerations	Dec 2022
Binghamton University Graduate Conference in Algebra & Topology	Nov 2022
AMS Sectional, Salt Lake City	Oct 2022
Young Topologists Meeting, Copenhagen	Jul 2022
Homotopical Methods in Fixed Point Theory, CU-Boulder	Jul 2022
Homotopy Theory with Applications to Arithmetic and Geometry, Fields Institute	Jun 2022
Algebraic K-theory, motivic cohomology and motivic homotopy theory, INI	Jun 2022
Algebraic Structures in Topology, San Juan	Jun 2022
Graduate Student Conference in Algebra, Geometry, & Topology, Temple University	May 2022
MIT Talbot Workshop: Ambidexterity in Chromatic Homotopy Theory, Plymouth MA	Oct 2021
Mathematics Teacher-Scholar Symposium, Reed College*	May 2021
Institute for Mathematics and Democracy 2021*	May 2021
Graduate Student Conference in Geometry and Topology*	Apr 2021
Midwest Topology Conference*	Apr 2021
Binghamton University Graduate Conference in Algebra and Topology*	Nov 2020
ICERM: Monodromy and Galois groups in enumerative geometry*	Aug-Sep 2020
Regensburg Transatlantic Transchromatic Homotopy Theory Conference II*	Aug 2020
IHES Motivic, Equivariant and Non-commutative Homotopy Theory*	July 2020
Motives and What Not, Universität Regensburg*	May 2020
Midwest Topology Seminar*	May 2020
MAGUS*	May 2020
Motives & Stacks, Universität Duisburg-Essen	Sep 2019
European Autumn School in Topology 2019, Utrecht	Sep 2019
AMS Fall Sectional, UW-Madison	Sep 2019
PIMS Workshop on Arithmetic Topology, UBC	Jun 2019
Graduate Student Conference in Algebra, Geometry, & Topology, Temple University	Jun 2019
Arizona Winter School 2019: Topology and Arithmetic	Mar 2019
Homotopy Theory Summer Berlin 2018	Jun 2018
Joint Mathematics Meeting, Atlanta	Jan 2017

SERVICE

Master TA, UPenn

In charge of running TA trainings and conducting observations.

The Franklin Institute Science Museum

2018 — 2022

Volunteer science presenter with Team Boson, responsible for running tables at exhibits and discussing science with the public.

Johns Hopkins Math Club

2015 — 2018

President (2017-2018), Vice President (2016-2017), organizer of speakers and events, test writer / grader / organizer for the Johns Hopkins Math Tournament, held yearly for high school students in the greater DC area. Founder of the Johns Hopkins Undergraduate Math Seminar.

EXPERIENCE

Noblis, Inc

Reston, VA

PhD intern in post-quantum cryptography.

Summer 2018

Texas A&M REU

College Station, TX

Analytic number theory, supervised by Dr. Matthew Young.

Summer 2017

Muhlenberg College REU

Allentown, PA

Number theory, supervised by Dr. Joshua Harrington.

Summer 2016